

● MEDIUM

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Inferences

10 questions · ~15 min

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Q1

INFORMATION AND IDEAS

Herbivorous sauropod dinosaurs could grow more than 100 feet long and weigh up to 80 tons, and some researchers have attributed the evolution of sauropods to such massive sizes to increased plant production resulting from high levels of atmospheric carbon dioxide during the Mesozoic era. However, there is no evidence of significant spikes in carbon dioxide levels coinciding with relevant periods in sauropod evolution, such as when the first large sauropods appeared, when several sauropod lineages underwent further evolution toward gigantism, or when sauropods reached their maximum known sizes, suggesting that _____blank

Which choice most logically completes the text?

- A. fluctuations in atmospheric carbon dioxide affected different sauropod lineages differently.
- B. the evolution of larger body sizes in sauropods did not depend on increased atmospheric carbon dioxide.
- C. atmospheric carbon dioxide was higher when the largest known sauropods lived than it was when the first sauropods appeared.
- D. sauropods probably would not have evolved to such immense sizes if atmospheric carbon dioxide had been even slightly higher.

Marine archaeologists have found much of the wooden hull of a sixteenth-century ship in a flooded quarry in southeast England. When it is exposed to air and water, wood rots quickly unless it is protected by sediment that shields it from oxygen. Therefore, the discovered ship was likely _____ blank

Which choice most logically completes the text?

- A. covered by a protective sediment layer in the quarry.
- B. one of several other ships buried in the same quarry.
- C. a confirmation of previous theories about the type of wood that was used in sixteenth-century ships.
- D. first constructed much earlier than previously thought.

Even with the widespread adoption of personal computers, many authors still choose to write and revise their novels by hand and only then transcribe the final version on a computer. It may be tempting to speculate about how a novel written this way would be affected if it had been exclusively typed instead, but each novel is a unique entity resulting from a specific set of circumstances. Therefore, _____ blank

Which choice most logically completes the text?

- A. in order to increase their efficiency, authors who currently write their novels largely by hand should instead work only on a computer.
- B. authors who do most of their drafting and revising by hand likely have more success than those who work entirely on a computer.
- C. novels written by hand take less time to produce, on average, than novels written on a computer do.
- D. there is no way to reasonably evaluate how a work would be different if it had been written by other means.

As the name suggests, dramaturges originated in theater, where they continue to serve a variety of functions: conducting historical research for directors, compiling character biographies for actors, and perhaps most importantly, helping writers of plays and musicals to hone the works' stories and characters. Performance scholar Susan Manning observes that many choreographers, like playwrights and musical theater writers, are concerned with storytelling and characterization. In fact, some choreographers describe the dances they create as expressions of narrative through movement; it is therefore unsurprising that _____ blank

Which choice most logically completes the text?

- A. dramaturges can have a profound impact on the artistic direction of plays and musicals.
- B. choreographers developing dances with narrative elements frequently engage dramaturges to assist in refining those elements.
- C. dances by choreographers who incorporate narrative elements are more accessible to audiences than dances by choreographers who do not.
- D. some directors and actors rely too heavily on dramaturges to complete certain research tasks.

Pigments give paints and dyes their color. Ocher is a mineral-based pigment used to make several colors, including red. Red ocher gets its color from iron oxide. Pigments can also be plant-based; plant-based pigments contain a high level of carbon. In a 2023 study, archaeologists tested the red pigment on decorated beads made by members of the Natufian culture approximately 15,000 years ago. The test showed that the pigment found on several beads contained no iron but had a high level of carbon. This finding led the researchers to conclude that _____ blank

Which choice most logically completes the text?

- A. the Natufian beads examined in the study are the oldest surviving examples of the use of plant-based pigments for decorating beads.
- B. the Natufian beadmakers used plant-based pigments rather than ocher to decorate some of the beads examined in the study.
- C. the Natufian beadmakers preferred to use plant-based pigments because they are much brighter than mineral-based pigments are.
- D. the pigments used by the Natufian beadmakers likely came from plants because ocher was difficult to find.

Some businesses believe that when employees are interrupted while doing their work, they experience a decrease in energy and productivity. However, a team led by Harshad Puranik, who studies management, has found that interruptions by colleagues can have a social component that increases employees' sense of belonging, resulting in greater job satisfaction that benefits employees and employers. Therefore, businesses should recognize that _____ blank

Which choice most logically completes the text?

- A. the interpersonal benefits of some interruptions in the workplace may offset the perceived negative effects.
- B. in order to maximize productivity, employers should be willing to interrupt employees frequently throughout the day.
- C. most employees avoid interrupting colleagues because they don't appreciate being interrupted themselves.
- D. in order to cultivate an ideal workplace environment, interruptions of work should be discouraged.

The Haitian Declaration of Independence was issued in 1804, bringing to an end the revolution against colonial France that began in 1791. Written in French, which was not the first language of most Haitians but which was used throughout Europe as the language of international diplomacy, the declaration notes that Haiti will not bring rebellion to other Caribbean nations, promises to respect the sovereignty of its neighbors—widely understood as a reassurance to the United States—and sets up Haiti as an example for future struggles against colonizers (an implicit reference to the many colonies then found in the Americas). So even though the declaration is explicitly addressed to the Haitian people, it's reasonable to conclude that _____ blank

Which choice most logically completes the text?

- A. aspects of the declaration were modeled on similar documents from other countries.
- B. the French government may have been surprised by the declaration.
- C. many Haitian people opposed the revolution and the declaration.
- D. the declaration actually had several intended audiences.

To investigate the history of plate subduction—when one of Earth’s tectonic plates slides beneath another—Sarah M. Aarons and colleagues compared ancient rocks from the Acasta Gneiss Complex in Canada to modern rocks. Using isotope analysis, the researchers found that Acasta rocks dating to about 4.02 billion years ago (bya) most strongly resemble modern rocks formed in a plume setting (an area in which hot rocks from Earth’s mantle flow upward into the crust). By contrast, they found that Acasta rocks dating to about 3.75 bya and 3.6 bya have an isotope composition that is similar to that of modern rocks formed in a subduction setting. Aarons’s team therefore concluded that _____ blank

Which choice most logically completes the text?

- A. subduction-like processes began occurring in some locations no later than 3.75 bya.
- B. subduction replaced mantle plume formation as the most common geological process by about 4.02 bya.
- C. the majority of the rocks in the Acasta Gneiss Complex formed through subduction.
- D. the rocks in the Acasta Gneiss Complex are of a more recent origin than scientists previously thought.

In documents called judicial opinions, judges explain the reasoning behind their legal rulings, and in those explanations they sometimes cite and discuss historical and contemporary philosophers. Legal scholar and philosopher Anita L. Allen argues that while judges are naturally inclined to mention philosophers whose views align with their own positions, the strongest judicial opinions consider and rebut potential objections; discussing philosophers whose views conflict with judges' views could therefore _____ blank

Which choice most logically completes the text?

- A. allow judges to craft judicial opinions without needing to consult philosophical works.
- B. help judges improve the arguments they put forward in their judicial opinions.
- C. make judicial opinions more comprehensible to readers without legal or philosophical training.
- D. bring judicial opinions in line with views that are broadly held among philosophers.

In a study of the mechanisms underlying associative memory—or the ability to learn and remember connections between inherently unrelated things—neuroscientists Kei Igarashi, Jasmine Chavez, and others presented mice with memory tests. The team discovered that fan cells, a type of cell found in the medial temporal lobe of the brain, are necessary for the acquisition of new associative memories. They also found that fan cell activity requires dopamine, a chemical the brain produces in response to pleasure and rewards. Consequently, receiving a reward should likely help to _____ blank

Which choice most logically completes the text?

- A. decrease an individual's capacity to utilize dopamine.
- B. increase an individual's capacity to recognize differences between unrelated things.
- C. increase an individual's capacity to form associative memories.
- D. decrease an individual's capacity to create fan cells.